

Experiment 14: Decision Tree

Aim

To implement a Decision Tree classifier using Python.

Objective

To classify data using a Decision Tree algorithm.

Algorithm

1. Import the required libraries.
 2. Prepare the dataset.
 3. Train the Decision Tree model.
 4. Predict the output.
 5. Display the prediction.
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Flowchart

Start

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Load Dataset

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Train Model

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Predict Output

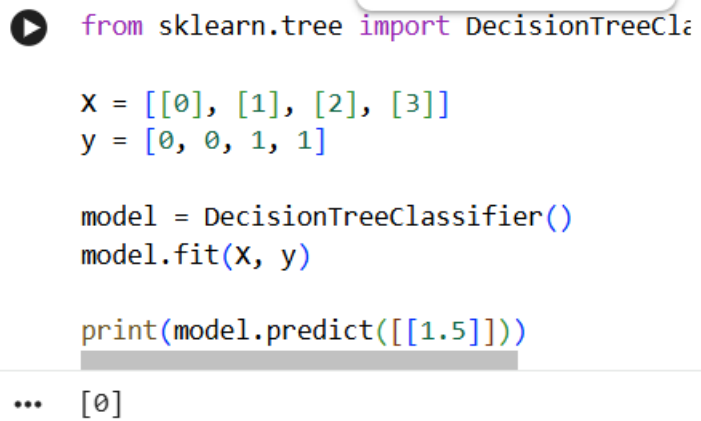
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Display Result

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Stop

Code and Output

A Jupyter Notebook code cell with a blue border and a play button icon. It contains Python code for training and predicting with a Decision Tree Classifier. The code defines feature matrix X and target vector y, creates a model, fits it, and predicts for a new input. The output is shown as [0].

```
from sklearn.tree import DecisionTreeClassifier

X = [[0], [1], [2], [3]]
y = [0, 0, 1, 1]

model = DecisionTreeClassifier()
model.fit(X, y)

print(model.predict([[1.5]]))
```

... [0]

Result

The Decision Tree model classified the given input successfully.

Conclusion

Decision Trees are simple and effective supervised learning algorithms used for classification tasks.